EcoRewind 2.0 — Restart Plan, Architectures & Publication Strategy

Oct 25, 2026 · @Kunal

Verdict: yes, restart it

EcoRewind is worth restarting. It is a genuinely uncommon student project, and it can become a first-author workshop paper and the strongest piece in your portfolio.

Why it isn't what every other student is doing. Most student remote-sensing work is land-cover classification, crop mapping or change detection. Those answer "what is here" or "what changed". EcoRewind asks a causal question: "what would this wetland look like today if the hurricane had never hit?" That puts it at the intersection of three fields that rarely meet in one student project: spatiotemporal deep learning, causal inference, and ecosystem science.

What is genuinely novel, and what isn't. Be precise about this in the paper, because reviewers will be.

- **Not new:** pixel-level counterfactual baselines already exist in remote sensing. Harmonic "what would this pixel be" models (the CCDC and BFAST family) and synthetic-control / CausalImpact-style designs are established. You must compare against them, not ignore them.
- **New enough:** learned *spatiotemporal* counterfactuals that use spatial context and SAR, evaluated with a rigorous falsification protocol (placebo tests, spatial blocking, external damage data).
- **Your strongest finding:** carbon attribution varied about 20× across backbones in v1. If that survives a clean redo, "model choice dominates attribution uncertainty" is a result the community needs to hear, and it holds whether or not your model wins.

Honest risks. The placebo test may show the counterfactual is unreliable. The two sites are a small sample for generalisation claims. Ground truth for "what would have happened" never exists, so every validation is indirect. The plan below is built so that each of these outcomes still produces a paper (see the last section).

What changes from v1. Leak-free spatial splits from day one; strong classical and causal baselines; deep models chosen for this data size; placebo and external validation before any carbon claim; and a compute plan that fits free GPUs.

Research question and paper structure

The paper asks: **can a model trained only on pre-disturbance satellite time series produce counterfactual trajectories reliable enough to attribute post-hurricane change, and how much does the attribution depend on the choice of model?**

Structure the paper around three claims, each with its own test. Each claim stands on its own, so a failure in one still leaves a paper.

Claim	Test that supports it	If it fails
C1. The counterfactual is accurate where no disturbance happened	Forecast skill on held-out spatial blocks and on placebo dates, vs classical baselines	Report which baseline wins; the benchmark is still a contribution
C2. The counterfactual gap detects real damage	Gap is near zero in placebo tests and agrees with independent damage data	Paper becomes the cautionary study (C3 alone)
C3. Attribution is sensitive to model choice	Spread of attributed loss across 6+ models, with uncertainty	This is the headline either way

Drop carbon from the core claims. Report it only as an illustrative conversion of attributed vegetation loss, using published carbon densities with ranges. Carbon in v1 inherited all the attribution uncertainty and multiplied it.

Data inventory and audit

You have about 31 GeoTIFFs per site (Mississippi delta for Hurricane Ida, Everglades for Hurricane Irma), each roughly 100 MB, so about 6 GB in total. That is small enough for a laptop and a free Kaggle dataset. v1 used 32 quarterly composites (2016 Q1 to 2023 Q4), so check first which quarter is missing per site.

Run this audit as week 1's first script and save the output as `reports/data_audit.md`:

1. **Completeness.** List quarters present per site; confirm whether a quarter is missing or merged, and whether both sites share the same calendar.
2. **Geometry.** CRS, pixel size (should be 10 m), extent, and pixel alignment across all files of a site. One misaligned file silently corrupts every time series.

3. **Bands.** Band order and names per file. v1 used Blue, Green, Red, NIR, NDVI, NDWI, SAR VV. Confirm whether SWIR or VH are present; they help (below).
4. **Value ranges.** Reflectance scale (0–1 or 0–10000), SAR in dB or linear, nodata value. v1's worst bug was a fill value that decoded to NDVI = −1.
5. **Coverage.** Per-quarter valid-pixel fraction for optical and SAR. v1 had SAR valid on only about 13% of evaluated pixels; find out why before building on it.
6. **Event timing.** Irma landfall was September 2017 (2017 Q3) and Ida was August 2021 (2021 Q3). Record exactly which composites are pre- and post-event per site.

Re-export if needed. If SWIR (B11, B12) or SAR VH are missing, re-export from Google Earth Engine now, and commit the export script this time. SWIR is the most sensitive optical signal for vegetation moisture and salt stress, and VH is more sensitive to vegetation structure than VV. Also consider adding a per-quarter cloud-free observation count band, since composites built from one clear image are less trustworthy than ones built from ten.

Audit outcome and decisions (2026-10-26)

The audit (`data_audit_final.md`) found that **both sites must be re-exported** before training. Decisions:

Finding	Decision
Mississippi v1 is open water at the delta mouth (median NDVI about −0.2), not wetland	Drop it; Mississippi = v2 (Barataria Bay marsh) only
SAR valid in only 4–5 quarters per site (each quarter 0% or ~100%): an export-filter bug	Re-export Sentinel-1 IW, both orbits, VV + VH
All files in EPSG:4326 with non-square ~8.7–9.0 × 9.9 m pixels	Export in UTM (17N Everglades, 15N Mississippi) at native resolution
Everglades has 6 pre-event quarters, one interpolated (2017Q2)	**Option 2:** add a Harmonized Landsat Sentinel-2 (HLS) 30 m track from 2013 for both sites
Event quarters are mostly pre-landfall (Q3 = Jul–Sep)	Export pre- and post-landfall partial composites; first full post-event quarter is Q4

Finding	Decision
Thin composites; Mississippi 2016Q2 had zero scenes	Per-pixel cloud masking instead of a scene-level cloud filter; add clear-observation counts
No SWIR or VH; NDVI = −1 and DN 0 fills	Add B11, B12, VH; recompute indices from bands; treat fills as invalid

Two data tracks:

- **Track A (Sentinel-2, 10 m), 2015Q3–2023Q4.** Primary track for Ida; one consistent Sentinel-2 product for the whole series.
- **Track B (HLS, 30 m), 2013Q2–2023Q4.** Primary track for Irma (about 17 pre-event quarters), and a resolution-sensitivity comparison for Ida.

Sentinel-1 SAR exists from late 2014, so Track B quarters before then carry a SAR mask.

Preprocessing and tensor building

All of this is CPU work. Do it locally, never on a paid GPU.

1. **Load and stack** each site into one array of shape (T, C, H, W) with rasterio, plus a validity mask of the same shape. Store as Zarr or chunked NumPy so later steps read tiles, not whole files.
2. **Validity masks, per band.** Optical valid where not cloud/shadow/nodata; SAR valid where backscatter is finite and nonzero. Never fill gaps before masking. Every loss and metric is computed on valid pixels only.
3. **Normalisation fitted on training data only.** Compute statistics from pre-event quarters in training blocks only. Store them in a JSON file with the data hash. NDVI and NDWI stay in physical units (−1 to 1); do not shift them.
4. **Gap handling.** Feed the mask to the model as an input channel and fill gaps with a learned token or the pixel's training-period climatology. Never fill with a constant that has a physical meaning.
5. **Static context channels (optional but cheap).** Elevation (Copernicus DEM), distance to coast, and a land-cover class. These help the model know that marsh and mangrove behave differently.
6. **Ecosystem masks.** Restrict evaluation to the wetland classes the paper is about, using an existing land-cover product, so open water and urban pixels do not inflate scores.

Leak-free spatial-block design

This is the single most important change from v1, where 50%-overlapping patches split at random made R^2 0.901 an upper bound.

- Divide each site into a grid of **large blocks** (for example 2.56 km × 2.56 km, which is 256 × 256 pixels). Assign whole blocks to folds.
- Leave a **buffer** of at least one patch width between blocks in different folds, and drop buffer pixels from evaluation.
- Use **3 spatial folds per site**, stratified so each fold contains a similar mix of damaged and undamaged area.
- Sample training patches at **stride 128 inside a block** (no overlap across folds; overlap within a training block is fine).
- Add a **cross-site test**: train on one site, test on the other. Even if it scores poorly, it tells reviewers how far the model generalises.

Temporal design

- **Counterfactual model:** trained only on quarters strictly before the event. Input 4–8 pre-event quarters, forecast the post-event quarters autoregressively or directly.
- **Placebo windows:** pretend a hurricane happened in earlier, undisturbed years and run the identical pipeline. Reserve these windows from training (details in Validation).

Model ladder

Build models from cheapest to most expensive, and do not move up a rung until the rung below is evaluated. v1 jumped straight to five deep models with no classical baseline; reviewers in remote sensing will ask first how you beat a harmonic regression.

The ladder serves two purposes: finding the best counterfactual (claim C1) and measuring attribution spread across model families (claim C3). Eight models from four families gives C3 real weight.

Rung	Model	Family	Compute	Why it's here
0	Persistence (last pre-event quarter)	Naive	CPU, minutes	Floor every model must beat
0	Per-pixel seasonal climatology	Naive	CPU, minutes	Captures seasonality only

Rung	Model	Family	Compute	Why it's here
1	Harmonic regression per pixel (trend + annual harmonics, CCDC-style)	Classical RS	CPU, under 1 hour	The domain-standard counterfactual; mandatory
1	Synthetic control / difference-in-differences with undisturbed control areas	Causal	CPU	The causal-inference reviewer's baseline
2	Gradient-boosted trees (LightGBM) on per-pixel lags, season, static context	Tabular ML	CPU, about 1 hour	Often beats deep models on small data
3	ConvLSTM	Spatiotemporal DL	GPU, short	Your v1 baseline, kept for continuity
3	SimVP or TAU	Spatiotemporal DL	GPU, short	Simple, strong, cheap video-prediction models
3	EcoTransformer v2	Spatiotemporal DL	GPU, medium	Your model, redesigned (below)
4	Presto-based head	Pretrained foundation model	GPU or CPU, short	Pretrained on Sentinel-1/2 pixel time series; lightweight

Verify each cited method's current paper and code before relying on it: SimVP (Gao et al., 2022), TAU (Tan et al., 2023), Presto (Tseng et al., 2023) and the CCDC/BFAST family. These are named from memory.

Why these and not more transformers

- **Small data favours inductive bias.** Two sites and about 30 timesteps is a small dataset. SimVP-style models and gradient-boosted trees tend to hold up better than large attention models at this size.

- **Pretrained models trade compute for data.** Presto was pretrained on large global Sentinel time series. Fine-tuning a small head on your data borrows that knowledge for almost no compute.
- **Classical baselines make deep results credible.** If EcoTransformer beats harmonic regression on held-out blocks, the result means something. If it doesn't, that's an honest finding too.
- **Optional, if a free GPU allows:** Earthformer (cuboid attention for Earth-system forecasting) or Prithvi-EO-2.0 (NASA/IBM geospatial foundation model). Both are heavier; add them only after the core ladder is done.

What to drop from v1

Drop UNet-Mamba, UNet-PatchTST and UNet-Temporal. With 4 input steps they had too few temporal tokens to work with. Report their v1 failure in one paragraph as a preliminary negative result; don't spend new compute diagnosing them.

EcoTransformer v2 design changes

1. **Direct multi-horizon output** instead of autoregressive rollout. v1's "winter dip" drift came from feeding predictions back in.
2. **Mask and static context as inputs**: the validity mask, elevation, distance to coast and land-cover class.
3. **Longer input window**: 8 quarters (two full seasonal cycles) instead of 4.
4. **Quantile outputs** (for example 10th, 50th and 90th percentiles) so every forecast carries its own uncertainty.
5. **Keep the SAR gate** but evaluate it with an ablation, since v1's SAR R^2 was negative.
6. **Simpler loss**: masked L1 or quantile loss first. Add SSIM or smoothness terms only if an ablation shows they help.
7. **Select checkpoints on validation R^2 per band**, never on loss alone. v1 showed loss can't detect a degenerate predictor.

Calibrated uncertainty for every model

- **Seeds as a deep ensemble.** Two to three seeds per deep model give an ensemble spread at no extra design cost.
- **Split conformal prediction.** Post-hoc, CPU-only, and model-agnostic. Calibrate intervals on a held-out calibration block so each model's 90% interval really covers about 90% of undisturbed pixels.

- **Report coverage**, not just interval width, for every model.

Validation protocol

No attribution number goes in the paper until it passes the falsification tests below. Run them in this order; the first two are mandatory.

1. Placebo in time (mandatory)

Pick 2–3 pre-event windows with no major disturbance (for example, a year before each hurricane). Train the counterfactual model only on data before each fake event date, then forecast the following quarters. The "attributed loss" should be near zero, and inside the model's uncertainty interval.

- Report placebo attributed loss with its interval, per model.
- **Pass criterion, declared before you run it:** for example, real-event attributed loss exceeds the 95th percentile of placebo losses. Write it into the repo before looking at results.

2. Placebo in space (mandatory)

After the real event, find areas the hurricane barely touched, using wind-field or storm-track data and distance from landfall. The counterfactual gap there should also be near zero. A model that shows "damage" everywhere is detecting its own drift, not the hurricane.

3. Agreement with independent damage evidence

Compare the attributed-loss map against independent sources. Check what's currently available before committing:

- **Louisiana (Ida):** the Coastwide Reference Monitoring System (CRMS) publishes field vegetation and site data across coastal Louisiana marshes.
- **Everglades (Irma):** published studies mapped Irma's mangrove damage with lidar and Landsat; the Florida Coastal Everglades LTER program also publishes field data.
- **Both:** NOAA storm surge and wind-field products as a damage-exposure proxy.

Use correlation or rank agreement with these sources, not pixel-perfect accuracy. Field data is sparse and measures different things.

4. Attribution sensitivity (claim C3)

For every model on the ladder, compute attributed vegetation loss (area-weighted NDVI or NDWI deficit) per site with conformal intervals. Then report:

- the spread across models, as a ratio and as a range;
- how much of that spread survives after excluding models that failed the placebo tests;
- whether the spread is smaller than v1's ~20×. That comparison is itself a finding.

5. Standard forecast metrics

Per band, per site, per spatial fold: R^2, RMSE and MAE on valid pixels only, plus interval coverage and width. Report a mean and spread across folds and seeds, never a single best run.

Compute plan

The core study fits in about **35–50 GPU-hours**. **Colab Pro is now the primary source:** 200 compute units (CCUs) a month for two months, about 400 in total. Kaggle runs in parallel, and RunPod drops to an emergency fallback, so the expected spend is **$0**.

Half the ladder (rungs 0–2 and all conformal calibration) runs on your laptop CPU. Only four models need a GPU.

Work	Runs	Hours per run (P100/T4 class)	GPU-hours
4 GPU models × 3 spatial folds × 2 seeds	24	0.5–1.5	~20–30
Placebo retrains: 2 windows × 4 models	8	0.5–1.5	~6–10
EcoTransformer v2 ablations (SAR gate, mask input, static context, loss terms)	4	~1.5	~6
Cross-site transfer: best 2 models × 2 directions	4	~1	~4
Debugging and reruns	—	—	~5

These times are estimates. Stride-128 sampling gives about 4× fewer patches than v1, and v1's runs took 4–9 hours on your college RTX A4000. Time your first EcoTransformer run and rescale.

Where the hours come from

1. **Colab Pro (primary).** Approximate burn rates: T4 about 1.5–2 CCUs/hour, L4 about 4–5, A100 about 11–15. Check the exact rate in Colab's resources panel. Use the T4 for most runs and the L4 when a model needs more memory; avoid the A100. The core study costs roughly 70–100 CCUs on T4s.
2. **Kaggle Notebooks (parallel).** Run seeds or baselines there while Colab trains another model, roughly doubling throughput.
3. **Your college**, if remote access to the A4000 is possible.
4. **RunPod (emergency only)** if Colab units run out before the two months end.

Colab rules: sessions disconnect on idle, on browser close and at time limits, and background execution may not be included in Pro, so check. Save checkpoints and metrics to Google Drive every epoch, and make every script resume. Check whether unused units expire, and plan to use them within the two months.

Avoid your internship's hardware unless your manager approves in writing.

Rules that keep it at the low end

- **Every script resumes from a checkpoint.** Kaggle sessions end; your runs shouldn't die with them.
- **Smoke-test on CPU first:** one epoch on ten patches, end to end, before any GPU run.
- **Kill bad runs early:** abort if validation NDVI R^2 is still negative after about 15 epochs.
- **Mixed precision (fp16 on T4/P100)** and the smallest model that works.
- **One config file per run, logged with the data hash,** so nothing has to be rerun because you can't tell what produced a number.

Expanded scope with Colab

The core plan plus SSA-Intel's GPU work uses about 130–290 of the ~400 CCUs. The rest funds a fuller EcoRewind study. Add these in priority order once the core results are in, and stop when units run low; keep about 40 CCUs in reserve for paper-revision reruns.

Priority	Addition	Approx. T4-hours	Approx. CCUs
1	5 spatial folds × 3 seeds for the 4 core GPU models (instead of 3 × 2)	+20–50	+40–100

Priority	Addition	Approx. T4-hours	Approx. CCUs
2	A third placebo window for every model	+5–10	+10–20
3	Earthformer and Prithvi-EO-2.0 on the ladder (L4 for memory)	+10–20 on L4	+45–90
4	A third site (download from Earth Engine on CPU first)	+10–20	+20–40

Priority 1 strengthens every claim. Priority 3 strengthens the attribution-sensitivity finding (claim C3) by adding two more model families. Priority 4 is what a journal such as *Remote Sensing of Environment* would want, so attempt it only if the workshop paper is already written.

Timeline

Plan on **12–14 weeks part-time** alongside your internship. Each phase ends with a gate; don't start the next phase until the gate passes.

1. **Weeks 1–2: data and foundations.** Data audit, re-export if SWIR/VH are missing, commit the Earth Engine script, build tensors, masks, spatial blocks and folds. Write tests for masking and leakage. *Gate: audit report done, leakage tests pass.*
2. **Week 3: CPU baselines.** Persistence, climatology, harmonic regression, synthetic control/DiD, LightGBM, all on the spatial folds. Build the evaluation harness (metrics, conformal intervals, attribution) once, used by every model. *Gate: baseline table complete.*
3. **Weeks 4–5: deep models.** ConvLSTM, SimVP/TAU, EcoTransformer v2, Presto head on Colab Pro, with Kaggle in parallel. One fold and one seed first, then all folds and seeds. *Gate: at least one deep model beats harmonic regression on held-out blocks, or you have documented that none does.*
4. **Week 6: placebo tests.** Placebo in time and space for every model. *Gate: this decides which paper you write (see the last section).*
5. **Week 7: external validation and attribution.** CRMS and Everglades damage comparisons, attribution spread across models, carbon as illustration only.
6. **Week 8: ablations and cross-site transfer.**
7. **Weeks 9–10: paper.** All tables and figures generated from code. Write, then get one outside reader (your guide or a CCAI mentor) to review.

8. **Weeks 11–12: portfolio demo**, built from saved predictions, and an arXiv preprint after submission.
9. **Weeks 13–14: slack.** Something will slip; it usually does.

Check deadlines now. Climate Change AI workshop deadlines are typically a few months before the host conference. If the next one is close, submit a Proposals-track paper from v1 results while v2 runs.

Publication strategy

Aim for a **Climate Change AI workshop paper first**, then extend to a journal. Confirm current tracks and deadlines on each venue's call for papers.

Order	Venue	What it needs	Fit
1	Climate Change AI workshop at NeurIPS or ICLR, Papers track	Leak-free results, baselines, placebo tests, honest limits	Best fit; 4-page format
1 (alt)	CCAI Proposals track	Idea, v1 results, validation plan	Use if the deadline arrives before v2 is done
2	Remote-sensing journals (for example *IEEE JSTARS* or *Remote Sensing*)	Full study, external validation, more depth	After the workshop
3	*Remote Sensing of Environment*	Strong external validation, more sites	Only with field-data agreement and 3+ sites

What makes reviewers say yes

- **Classical baselines in the main table**, including harmonic regression and synthetic control.
- **Spatial-block folds**, stated in the abstract.
- **Placebo results with pre-declared pass criteria.**
- **Uncertainty on every number**, with interval coverage reported.
- **Negative results stated plainly**, including v1's leakage and diverged models. This builds trust.

- **Code and processed data released** with a fixed commit, plus a Zenodo DOI.

Practicalities

- **Affiliation.** If v1 was done at SIT, confirm with your guide whether it should appear as your affiliation and whether they should be a co-author. Agree this before you write.
- **arXiv preprint** after submission, which also gives your portfolio a citable link.
- **CCAI mentorship.** CCAI has run a mentorship program pairing early-career authors with experienced researchers; apply if it's open.

Portfolio piece

The demo should let a visitor **rewind a hurricane** with a slider. It needs no GPU in production: precompute every prediction once, export map tiles, and serve a static site.

The demo

1. **Swipe map.** Pick a site and a quarter. Swipe between what the satellite actually saw and what the model says would have been there without the hurricane. The difference layer shows attributed damage.
2. **Time slider.** Scrub quarter by quarter from 2016 to 2023 and watch the observed and counterfactual trajectories diverge at landfall.
3. **Click any pixel** to see its full time series: observed values, the counterfactual median and its uncertainty band.
4. **Model switcher.** Swap between models and watch the attributed-damage map change. This makes claim C3 visible in seconds, and it's the most memorable part.
5. **Placebo toggle.** Show the same view on a fake event date, where the gap should be near zero. Visitors see you tested your own claim.

Build it with cloud-optimised GeoTIFF or PMTiles layers in MapLibre GL, hosted free on GitHub Pages or Cloudflare Pages. Pixel time series can be small precomputed JSON files per tile.

How to present it

- **Lead with the question**, not the architecture: "What would the Everglades look like if Irma had never happened?"
- **Show the falsification**: the placebo map earns more trust than any R^2.

- **Tell the v1 to v2 story briefly**: a leaky split and an inflated R^2 of 0.901, found and fixed by you. Hiring managers value that more than a clean first attempt.
- **Keep the existing SVG assets** in the wetland palette, updated with v2 numbers.

Repository, testing and reproducibility

Start a fresh repository; keep v1 as a tagged archive, linked from the README.

```
ecorewind/
  gee/            Earth Engine export script (committed this time)
  configs/        one YAML per experiment
  src/
    data/         loading, masks, normalisation, blocks and folds
    models/       baselines, deep models, Presto head
    eval/         metrics, conformal intervals, attribution, placebo
  scripts/        train.py, evaluate.py, make_figures.py
  tests/
  notebooks/      Kaggle entry points only
  paper/          LaTeX; tables and figures generated by scripts
  demo/           static site and tile export
```

Tests that matter most

- [ ] **Leakage:** no pixel appears in two folds, buffers are respected, normalisation stats come from training blocks only.
- [ ] **Masking:** a pixel marked invalid never affects a loss or metric.
- [ ] **Fill values:** no gap-fill value decodes to a physical extreme (the v1 NDVI = −1 bug).
- [ ] **Temporal cutoff:** the counterfactual model never sees a post-event or post-placebo quarter.
- [ ] **Metrics:** R^2, RMSE and conformal coverage checked against hand-computed fixtures.
- [ ] **Smoke run:** one epoch on ten patches runs end to end on CPU, in CI.

Reproducibility checklist

- [ ] One command regenerates every table and figure from saved predictions.
- [ ] Every run logs its config, git commit and data hash.
- [ ] Seeds fixed and listed.
- [ ] Pinned environment file.

- [ ] Processed data and checkpoints released (Zenodo or Hugging Face), within Copernicus data terms.

Risks and fallback papers

Every outcome of the placebo test still leads to a paper. Decide which one you're writing at the week-6 gate.

Outcome at week 6	Paper you write
Deep models beat baselines and pass placebo tests	**Method paper:** learned spatiotemporal counterfactuals for hurricane attribution, with falsification
Classical baselines match or beat deep models, and pass placebo	**Benchmark paper:** what actually works for wetland counterfactuals, and when deep learning isn't worth it
Most models fail placebo tests	**Cautionary paper:** learned counterfactual attribution is model-dependent and often unfalsifiable; here is a protocol to check it

Risk	Likelihood	Mitigation
Free GPU quota runs out	Medium	CPU baselines first; RunPod fallback of $15–30
SAR adds nothing	High	Report the ablation; keep SAR only if it helps
No usable external ground truth	Medium	Lean on placebo tests; frame field data as qualitative agreement
Two sites too few for generalisation	High	Claim nothing beyond the two sites; cross-site test as a limitation
Workshop deadline arrives early	Medium	Proposals track with v1 results

Kill criteria, declared now

- Stop training any model whose validation NDVI R^2 stays negative after about 15 epochs.
- Don't report carbon if attributed vegetation loss fails the placebo test.
- Don't add a new architecture after week 5. Write with what you have.